Quiz-1: DA526 Image Processing with Machine Learning (Jan-Apr 2026)

Date: 17-FEB-2026 **Maximum Marks: 45**

NAME: ______________________________ **ROLL NO.:** ______________________

Instructions:

- *There is total 18 questions. All questions are compulsory.*
- *For MCQs, select and fill only the correct option.*
- *Write your answers clearly in the space provided for the fill-in-the-blank questions.*
- *Use answer sheets for descriptive questions.*
- *Ensure that your name and roll number are written on question paper and answer sheets.*

Q1 [1 mark]. Brightness adaptation in human vision primarily refers to:

◯ (A) Ability to detect fine edges
◯ (B) Ability to perceive objects in varying illumination levels
◯ (C) Ability to increase image contrast artificially
◯ (D) Increasing bit depth of an image

Q2 [1 mark]. The human visual system can perceive brightness over:

◯ (A) A narrow fixed intensity range
◯ (B) Only linear intensity changes
◯ (C) A very wide dynamic range through adaptation
◯ (D) Only low illumination levels

Q3 [1 mark]. Spatial resolution of a digital image depends primarily on:

◯ (A) Number of gray levels
◯ (B) Number of pixels used to represent the image
◯ (C) Bit depth
◯ (D) Compression ratio

Q4 [1 mark]. If an image of size 256×256 is resized to 1024×1024 without adding new information, then the spatial resolution:

◯ (A) Increases
◯ (B) Decreases
◯ (C) Remains fundamentally unchanged
◯ (D) Becomes continuous

Q5 [1 mark]. Increasing intensity resolution while keeping spatial resolution fixed will:

◯ (A) Improve geometric details
◯ (B) Reduce image size
◯ (C) Reduce quantization error
◯ (D) Increase aliasing

Q6 [1 mark]. Quantization primarily affects:

◯ (A) Spatial coordinates
◯ (B) Intensity values
◯ (C) Image size
◯ (D) Sampling rate

Q7 [1 mark]. Dynamic range of an imaging system is defined as:

◯ (A) Ratio of maximum to minimum measurable intensity
◯ (B) Total number of pixels
◯ (C) Sampling frequency
◯ (D) Image dimensions

Q8 [1 mark]. Increasing both spatial and intensity resolution will:

◯ (A) Always improve perception
◯ (B) Increase storage requirements
◯ (C) Reduce computational cost
◯ (D) Reduce noise automatically

Q9 [1 mark]. Increasing polynomial degree p primarily increases:

◯ (A) Bias
◯ (B) Variance
◯ (C) Irreducible noise
◯ (D) Sample size

Q10 [2 marks]. The gradient of $f(w) = \frac{1}{2}\|Xw - y\|^2$ is ____________________.

Q11 [1 mark]. Least squares require $X^T X$ to be ____________________ for invertibility.

Q12 [2 marks]. All eigenvalues of covariance matrix are _____________________ and _____________________.

Q13 [1 mark]. High bias corresponds to model being too _____________________.

Q14 [4 marks]. Explain why high-dimensional data causes distance measures to become less meaningful.

Q15 [4 marks]. Explain why adding bias term improves linear model. Rewrite least squares formulation including bias.

Q16 [7 marks]. Consider the standard Digital Image Processing pipeline. Draw the fundamental steps and explain how low-level, mid-level, and high-level processes are represented within this framework.

a b c

FIGURE 2.25 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Q17 [5 marks]. The three images shown in Figure 2.25 (a), (b), and (c) show scenes with increasing levels of detail. Suppose a constant value k is added to every pixel as:

$$g(x,y)=f(x,y)+k$$

If k is large, list FOUR possible visual or technical effects that may occur in the images.

Q18 [10 marks]. In continuation of **Q17**, consider reducing intensity resolution of the images from 8-bit (256 levels) to 3-bit (8 levels), while keeping spatial resolution unchanged.

- **(i)** List FOUR effects this reduction may produce.
- **(ii)** Also briefly indicate which image (a), (b), or (c) will be most affected and why.